

DESIGN AND SIMULATION OF A 4-BIT BINARY-TO-DECIMAL CONVERSION AND DISPLAY SYSTEM

DDCA Project | Digital Logic Design | Electronics and Communication Engineering

PROJECT TEAM

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INSTITUTION DETAILS

Department: Electronics and Communication Engineering

College: KLH

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INTRODUCTION

This project delves into the core principles and practical implementation of 4-bit binary-to-decimal conversion and display systems.

- **4-BIT BINARY NUMBERS**

Represent values from 0000 to 1111, covering decimal equivalents from 0 to 15.

- **BINARY-TO-DECIMAL CONVERSION**

The process of interpreting a binary sequence to reveal its equivalent decimal numeric value.

- **DIGITAL LOGIC SYSTEMS**

Utilize combinational circuits to efficiently process binary inputs and generate meaningful outputs.

- **DISPLAY SYSTEMS**

Crucial for converting abstract digital signals into human-readable formats, such as LEDs or seven-segment displays.

- **PRACTICAL RELEVANCE**

Understanding these concepts is foundational for digital electronics, microcontroller interfacing, and modern embedded systems design.

PROBLEM STATEMENT: BRIDGING BINARY AND HUMAN UNDERSTANDING

The fundamental challenge in digital systems often lies in translating machine-level binary data into a format that is intuitive and usable for humans. Our project addresses this critical interface.

- **THE BINARY INTERPRETATION GAP**

A 4-bit binary input, such as 1010, is a sequence of bits that lacks inherent meaning for most users, requiring complex mental conversion to understand its decimal equivalent (10).

- **HUMAN-CENTRIC DATA REPRESENTATION**

Humans primarily operate within a decimal system, making direct binary representations inefficient and prone to error in user interfaces and readouts.

- **ESSENTIAL DIGITAL CONVERSION**

Digital systems require robust conversion logic to transform binary inputs into decimal outputs for practical display and seamless user interaction across various devices.

- **FOUNDATIONAL EDUCATIONAL INSIGHT**

Designing a binary-to-decimal converter provides invaluable hands-on experience for students, illuminating core concepts of combinational logic, truth tables, and decoder circuits.

- **SIMULATION FOR VALIDATION**

Utilizing simulation tools like Logisim offers a safe, cost-effective, and iterative environment to test and refine circuit designs before committing to physical implementation.

OBJECTIVES

1. Accept a 4-bit binary input from the user or input pins
2. Convert the binary value into its decimal equivalent (0–15)
3. Design the required combinational logic using basic gates and decoders
4. Generate appropriate decimal or BCD output signals
5. Display the resulting decimal value using a seven-segment display or LED indicators
6. Simulate the complete circuit in Logisim
7. Test all 16 possible input combinations and verify correct decimal outputs
8. Document the truth table and verify against expected results

LITERATURE SURVEY & EXISTING SYSTEMS

A review of established methods and technologies forms the foundation for our design, highlighting both effective strategies and common challenges.

BINARY-TO-DECIMAL CONVERSION METHODS

Primarily based on positional notation (e.g., summing 2^3 , 2^2 , 2^1 , 2^0 for a 4-bit input) or decoder-based approaches using a 4-to-16 line decoder.

- **DISPLAY TECHNOLOGIES**

Commonly utilize seven-segment displays requiring BCD-to-seven-segment decoder logic, or simple LED arrays for direct bit visualization.

- **BINARY-CODED DECIMAL (BCD)**

A system encoding each decimal digit separately, beneficial for multi-digit displays but introduces conversion complexity.

- **LIMITATIONS OF EXISTING APPROACHES**

Complex multi-digit systems demand extensive additional logic, hardware decoders contribute to overall cost, and direct hardware implementation can limit educational insights without simulation tools.

PROPOSED SYSTEM

Our proposed system is a Logisim-based 4-bit binary-to-decimal converter, designed to offer a clear, interactive, and educational solution for digital data representation.

SYSTEM OVERVIEW

Converts 4-bit binary input to equivalent decimal value (0-15) for display.

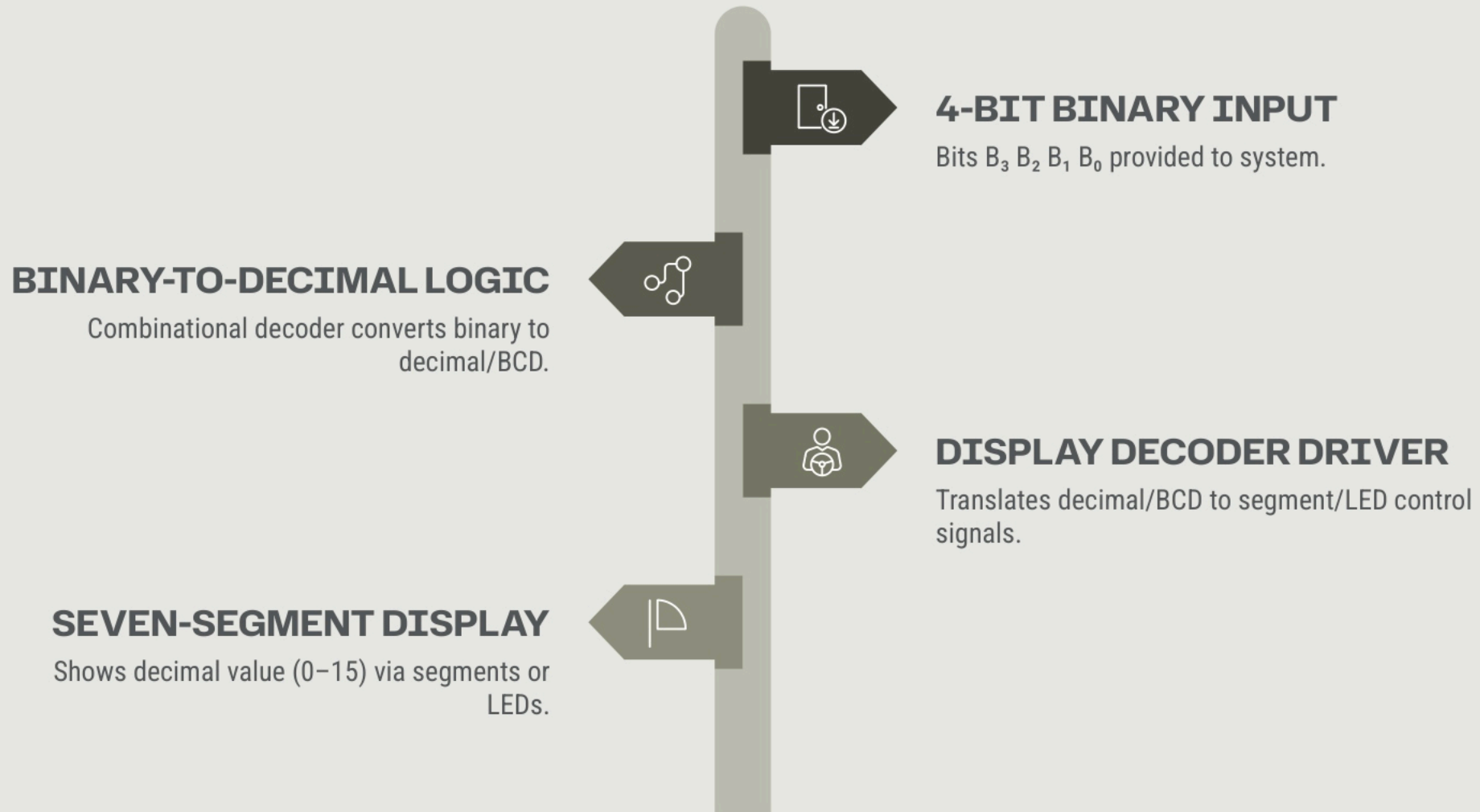
KEY COMPONENTS

- 4-bit binary input (switches)
- Combinational logic (gates/decoder)
- Decimal output representation
- Seven-segment display/LED indicators

ADVANTAGES

- Clear visualization of binary-to-decimal mapping
- Easy testing of all 16 input combinations
- Safe, cost-free simulation environment
- Reinforces logic & decoder understanding

SYSTEM ARCHITECTURE / BLOCK DIAGRAM



METHODOLOGY / IMPLEMENTATION

1. **Input Stage:** Define four binary input bits (B_3, B_2, B_1, B_0) in Logisim using input pins or switches
2. **Conversion Logic Design:** Create combinational logic using basic AND, OR, NOT gates to implement the conversion function, or use a 4-to-16 decoder IC to map each input combination to a unique output line
3. **Output Generation:** Generate decimal value (0–15) or BCD representation from the conversion logic
4. **Display Connection:** Connect the output to a seven-segment display decoder (or LED array) to visualize the result
5. **Circuit Assembly:** Build the complete circuit in Logisim, connecting all components with proper signal routing
6. **Testing:** Systematically apply all 16 input combinations (0000 to 1111) and observe the displayed output
7. **Verification:** Compare displayed values against the expected truth table to confirm correct operation

DESIGN APPROACHES

Comparing two implementation strategies for the binary-to-decimal converter circuit.

APPROACH 1: GATE-LEVEL IMPLEMENTATION

- Uses basic AND, OR, NOT gates
- Requires deriving Boolean expressions from truth table
- More gates needed, but demonstrates fundamental logic design
- Educational value: reinforces Karnaugh maps and Boolean algebra
- Complexity: Medium (16 expressions for 16 outputs)

ADVANTAGES:

- Deep understanding of fundamental logic
- Customizable and flexible circuit design

DISADVANTAGES:

- Higher component count
- More complex wiring
- Increased potential for wiring errors

APPROACH 2: DECODER-BASED IMPLEMENTATION

- Uses a 4-to-16 decoder Integrated Circuit (IC)
- Direct mapping: each input combination activates one unique output line
- Simpler circuit architecture, fewer connections
- Educational value: introduces decoder concept and abstraction
- Complexity: Low (single component + display logic)

ADVANTAGES:

- Significantly simpler design
- Fewer interconnections
- Faster implementation and debugging

DISADVANTAGES:

- Less direct insight into gate-level logic
- Reduced flexibility for custom logic modifications

PERFORMANCE ANALYSIS & VERIFICATION

Our rigorous analysis ensures the binary-to-decimal converter meets all functional and operational requirements.



CIRCUIT PERFORMANCE METRICS

- Propagation Delay: Time for signal travel.
- Power Consumption: Minimal in simulation.
- Gate Count: Total logic gates used.



VERIFICATION STRATEGY

- Systematic testing of all 16 input combinations.
- Comparison against expected truth table.
- Timing analysis for race conditions.



SIMULATION RESULTS

- All 16 inputs tested successfully.
- Output matched expected decimal values.
- No timing violations or glitches observed.



RELIABILITY ASSESSMENT

- 100% accuracy across all test cases.
- Consistent performance across multiple runs.
- Ready for hardware implementation.

RESULTS AND DISCUSSION - TRUTH TABLE

Binary Input	Decimal Output
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	10
1011	11
1100	12
1101	13
1110	14
1111	15

The circuit was simulated in Logisim. All 16 input combinations were tested, and the displayed decimal outputs matched the expected truth table values, confirming correct operation of the binary-to-decimal conversion logic.

CONCLUSION

Our project successfully demonstrated the design, simulation, and verification of a binary-to-decimal converter.



PROJECT SUMMARY

Designed and simulated a 4-bit binary-to-decimal conversion and display system using Logisim.



KEY ACHIEVEMENT

The system accurately converts all 16 possible 4-bit binary inputs (0000–1111) into their decimal equivalents (0–15).



VERIFICATION

Rigorous simulation testing confirmed correct operation across all input combinations, ensuring high accuracy.



LEARNING OUTCOMES

- Understanding of combinational logic and truth tables
- Practical knowledge of decoder circuits and applications
- Experience with digital display systems
- Hands-on circuit design and simulation in Logisim



FUTURE SCOPE

- Extend to multi-digit decimal displays (8-bit or 16-bit inputs)
- Implement BCD encoding for more complex systems
- Hardware implementation using breadboard or FPGA
- Add real-time input switching and visual feedback